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Real-Time Proprioceptive Sensing Enhanced Switching Model Predictive Control for Quadruped Robot Under Uncertain Environment

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Abstract

Quadruped robots have shown significant potential in disaster relief applications, where they have to navigate complex terrains for search and rescue or reconnaissance operations. However, their deployment is hindered by limited adaptability in highly uncertain environments, especially when relying solely on vision-based sensors like cameras or LiDAR, which are susceptible to occlusions, poor lighting, and environmental interference. To address these limitations, this paper proposes a novel sensor-enhanced hierarchical switching model predictive control (MPC) framework that integrates proprioceptive sensing with a bi-level hybrid dynamic model. Unlike existing methods that either rely on handcrafted controllers or deep learning-based control pipelines, our approach introduces three core innovations: (1) a situation-aware, bi-level hybrid dynamic modeling strategy that hierarchically combines single-body rigid dynamics with distributed multi-body dynamics for modeling agility and scalability; (2) a three-layer hybrid control framework, including a terrain-aware switching MPC layer, a distributed torque controller, and a fast PD control loop for enhanced robustness during contact transitions; and (3) a multi-IMU-based proprioceptive feedback mechanism for terrain classification and adaptive gait control under sensor-occluded or GPS-denied environments. Together, these components form a unified and computationally efficient control scheme that addresses practical challenges such as limited onboard processing, unstructured terrain, and environmental uncertainty. A series of experimental results demonstrate that the proposed method outperforms existing vision- and learning-based controllers in terms of stability, adaptability, and control efficiency during high-speed locomotion over irregular terrain.

Keywords: quadruped robot; proprioceptive sensing; hybrid dynamic modeling; switching-based model predictive control; situational awareness

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Academic Editor: Maciej Ławryńczuk

Received: 2 May 2025
Revised: 16 June 2025
Accepted: 23 June 2025
Published: 2 July 2025

Citation: Lokhande, S.; Bao, Y.; Cheng, P.; Shen, D.; Chen, G.; Xu, H. Real-Time Proprioceptive Sensing Enhanced Switching Model Predictive Control for Quadruped Robot Under Uncertain Environment. *Electronics* **2025**, *14*, 2681. https://doi.org/10.3390/electronics14132681

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1. Introduction

Special autonomous robotics systems (e.g., legged robots) have recently been successfully used in various reconnaissance missions. In 2009, DARPA started a legged squad support system (LS3) project to develop a legged robot that can function autonomously as a pack horse for a squad of soldiers or marines. Under the LS3 project, a series of quadruped robotic systems have been developed in the U.S., including "Spot," "Mini-Spot" from

Boston Dynamics, "Vision 60," "Spirit 40" from Ghost Robotics, "Cheetah," "Mini-Cheetah" from MIT, etc. Due to the substantial potential for future search and rescue missions, quadruped robotic development has also attracted international attention. It includes "ANYmal" from ANYbotics in Switzerland, "BEX" from Kawasaki Heavy Industries in Japan, "Go 1" from Unitree Robotics in China, and so on. However, those platforms with their out-of-the-box controllers have limited adaptability for challenging environments such as highly uneven, obstructed, and uncertain terrain [1,2]. Also, all existing quadruped robots suffer from limited agility, primarily due to high computational complexity that is often unaffordable for onboard systems [3,4] and slow mobility caused by inefficient real-time sensing capability [5,6]. Several control strategies have been developed to improve the locomotion of quadruped robots over uncertain terrains. Model predictive control (MPC) has emerged as a promising solution to incorporate constraints and optimization objectives directly in the control formulation [7]. It has been widely applied in quadruped locomotion for its ability to handle multi-variable constraints and anticipate future states based on the given model. Early implementations, such as Di Carlo et al. [8], demonstrated convex MPC formulations on MIT's *Cheetah 3*, enabling fast and efficient trajectory planning. These works, however, rely on simplified centroidal dynamics and assume relatively structured environments. Subsequent research [9] proposed Whole body control (WBC) and whole body impulse control (WBIC) [7,10] to improve stability and dynamic consistency across different contact modes. In their implementation, traditional MPC is used to generate the center of mass (CoM) trajectory, whereas WBC focuses on following the generated CoM trajectory by manipulating the actuators and joint positions. This approach presents a viable way, as it provides a dynamically consistent formulation that makes it scalable to more complex systems as well as varied tasks; however, they are not effective in providing enough computations for frequent non-contact phases as required during high-speed running modes. To address this, ref. [11] formulate WBC to follow both reaction forces and CoM trajectories by incorporating pre-computed reaction forces given by MPC as desired reaction forces and relaxing the floating base control. But this strategy is still inefficient when unstructured and uneven terrain operation is considered.

Alternative approaches—particularly learning-based methods—have demonstrated compelling results in challenging real-world conditions. For example, ref. [12] proposed a reinforcement learning-based control pipeline for the ANYmal robot that enabled agile navigation through complex tasks such as stair climbing, gap crossing, and even parkour-style maneuvers. Nonetheless, their dependence on extensive training data, high computational demand, a clear view of foothold positions, and limited generalizability to out-of-distribution terrains remain key drawbacks in practical deployment. Despite recent advances, current quadruped control architectures lack an integrated framework that can simultaneously handle hybrid dynamic modeling, terrain uncertainty, and real-time control with minimal sensing dependencies. This gap becomes critical in field deployments where exteroceptive sensors are unreliable or unavailable. Thus, there is a need to identify and recognize the terrain at low computational costs before utilizing model predictive control for a quadruped robot working in an unstructured setting. Researchers in [13] provided a simple and inexpensive way of classifying the terrain using only the time series data from the actuator. However, to further use this valuable information in a hierarchical way such that the optimization objective is solved at each time step has still not been addressed as evident in the literature survey of articles from [14–35].

This work aims to address these gaps by proposing a novel, proprioception-enhanced switching control framework. Our approach combines a bi-level hybrid dynamic model with a three-layer control architecture that integrates real-time IMU-based proprioceptive sensing, switching MPC for gait and trajectory adaptation, and distributed control for

leg stability. This framework is designed to enhance adaptability, reduce computational complexity, and enable robust locomotion across a wide range of unstructured terrains at high speed. With this, the following challenges are addressed:

- Lacking effective modeling techniques that consider the terrain and robotic limb reaction forces together.
- Lacking robust control that provides quadruped robot stability in discontinuous events caused by uncertain terrains.
- Lacking effective framework that considers both the modeling and control in a unified framework.

The development and validation of control strategies are elaborated here that can leverage both the simplified torso dynamics and the detailed leg dynamics by real-time feedback from a network of connected inertial measurement units (IMUs) placed at key locations of the quadruped robot skeleton. The new type of proprioceptive sensing-assisted control can improve the robot's stability, adaptability, and overall performance even in challenging environments. Specifically, this paper introduces:

i Situation-aware Hierarchical Bi-level Hybrid Dynamic Modeling: This strategy reduces the model complexity by integrating kinetic dynamics and multi-body dynamics hierarchically, thereby enhancing model agility by representing the multi-body dynamics in a distributed interconnected manner, and strengthens model scalability by incorporating a situation-driven multi-modal switched system modeling.

ii Hierarchical Hybrid Switching Control Framework: This strategy uses a three-layer hybrid control scheme that includes switching model predictive control to optimize the moving trajectory and gait schedule, robust distributed impulse torque control to enhance robotic leg robustness without increasing computational complexity, and fast-switched proportional derivative (PD) control to enhance the adaptability of quadruped robots.

Figure 1 shows the control architecture, and Figure 2 shows the overall framework. Both of these depict the data flow model that operates in a hierarchy. IMU-based proprioceptive state estimates, CoM trajectory, and the robot actuator states flow into the hybrid modeling and switched controller, and the solution of the co-joint algorithm feeds back into the quadruped robot, and the cycle continues until the goal is reached.

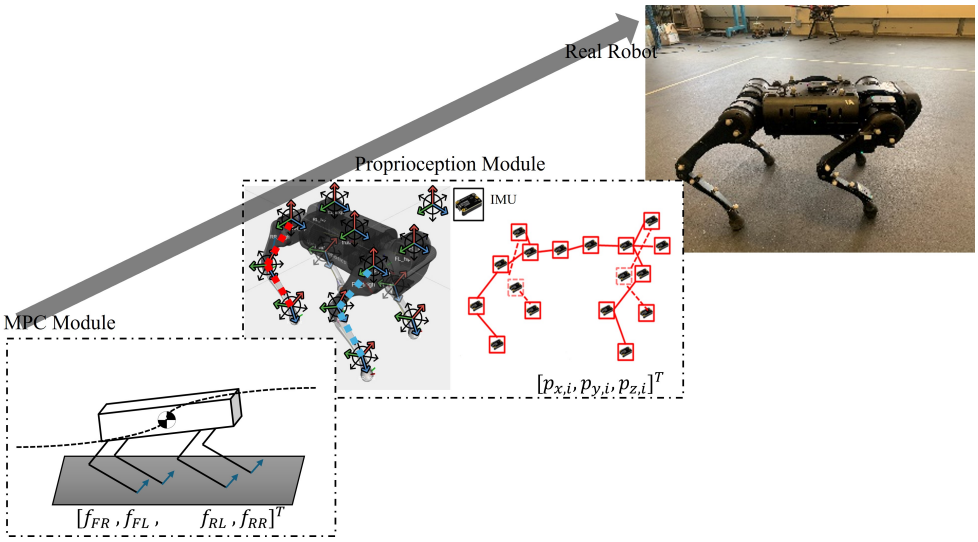


Figure 1. Control Architecture.

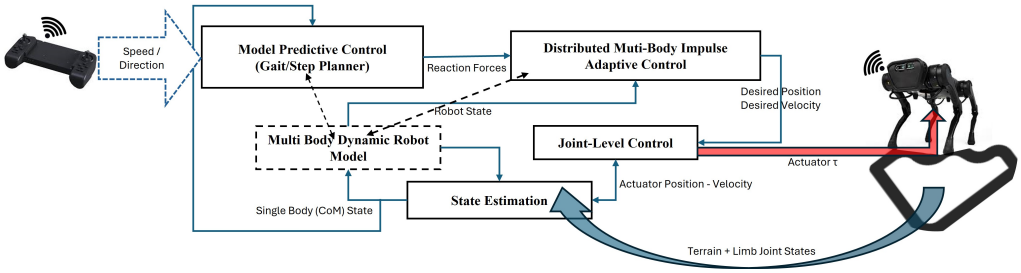


Figure 2. Overall Framework.

2. Methods

2.1. IMU-Based Proprioception Sensing

Proprioceptive sensing refers to the capability of biological creatures to sense the relative position of their bodies and limbs without relying on visual cues. This sense is critical for performing complex movements, maintaining balance, and reacting to unexpected changes in the environment. In robotic systems, proprioception is equally important, particularly for legged robots operating in harsh environments where external sensors like cameras or LiDAR may be obstructed, impaired by dust, rain, or darkness, or simply insufficient to provide the detailed information needed for precise control. Uneven terrains present unique challenges that make proprioceptive sensing not just beneficial but necessary for effective locomotion. Recent works, such as Chen et al. [36], have shown that proprioceptive sensing can significantly improve energy efficiency and adaptability through optimized foothold planning and posture adjustment. These findings support the use of IMUs and internal sensors as a reliable alternative or complement to exteroceptive systems for terrain-adaptive locomotion. IMUs are efficient in providing instantaneous movement information but lack long-term tracking information due to inherent biases. This makes these sensors suitable for implementing them as a key proprioceptive tool. With the rise of computationally cheap off-the-shelf inertial measurement units researchers [16] have been able to include multiple IMU units to estimate the traversed trajectory of the robots. However, they have not used an IMU to complement the proprioceptive sensing information. This paper introduces strategic placement of these IMUs on key joints of a four-legged robot that can enhance the robot's proprioceptive capabilities as depicted in Figure 3. Each IMU sensor provides localized data that, when integrated, offers a comprehensive understanding of the robot's overall posture and movement dynamics, leading to a well-versed robotic dynamic model.

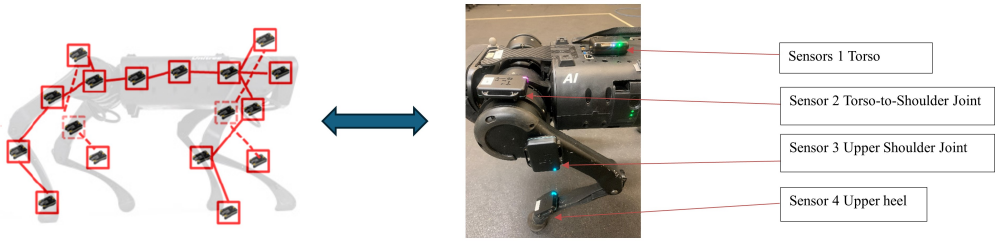


Figure 3. Strategic placements of IMU sensors on a quadruped.

The challenges of modeling the quadruped robotic system lie in the trade-off between modeling accuracy and modeling complexity and the conflicts between limited onboard sensors and uncertain, challenging terrains. To overcome the challenges, this paper develops a model to reformulate the quadruped robot model hierarchically to enable parallel computing and use simple distributed proprioception sensors to enable enhanced

state estimation. A specified form of the distributed multi-body dynamic model for the quadruped robot is shown in Figure 3. Inertial measurement units (IMUs) were selected for proprioception due to their advantages at low energy cost, low latency, and easy integration with other sensors.

Each leg is modeled separately as shown below:

$$\begin{aligned}
\text{Leg 1:} \quad & \dot{x}_1(t) = f_1(x_1) + g_1(x_1)u_1(t) + h_{s1}(x_1)s_1(t);\\
\text{Leg 2:} \quad & \dot{x}_2(t) = f_2(x_2) + g_2(x_2)u_2(t) + h_{s2}(x_2)s_2(t);\\
\text{Leg 3:} \quad & \dot{x}_3(t) = f_3(x_3) + g_3(x_3)u_3(t) + h_{s3}(x_3)s_3(t);\\
\text{Leg 4:} \quad & \dot{x}_4(t) = f_4(x_4) + g_4(x_4)u_4(t) + h_{s4}(x_4)s_4(t);\\
\text{Front Body:} \quad & \dot{x}_F(t) = f_F(x_F) + g_F(x_F)u_F(t) + h_{sF1}(x_F)s_{F1}(t) + h_{sF2}(x_F)s_{F2}(t);\\
\text{Back Body:} \quad & \dot{x}_B(t) = f_B(x_B) + g_B(x_B)u_B(t) + h_{sB1}(x_B)s_{B1}(t) + h_{sB2}(x_B)s_{B2}(t);
\end{aligned}$$

These equations are further simplified as

$$\begin{aligned}
z_1(t) &= h_1(x_1(t))\\
z_2(t) &= h_2(x_2(t))\\
z_3(t) &= h_3(x_3(t))\\
z_4(t) &= h_4(x_4(t))
\end{aligned}$$

$$z_F(t) = h_F(x_F(t)) \qquad z_B(t) = h_B(x_B(t)) \tag{1}$$

where $z_i(t) = h_i(x_i(t))$ denotes the output of subsystem i, where z_i represents the coupling output and $h_i(.)$ is the corresponding output function for leg $i = 1, 2, 3, 4$.

Similarly, $z_F(t) = h_F(x_F(t))$ and $z_B(t) = h_B(x_B(t))$ represent the output of the front and back subsystems, respectively. x represents subsystem state, u represents subsystem input, and s represents coupling input.

Unlike existing research for state estimation using IMUs, this subsection develops a novel unified two-layer deep mathematical model that thoroughly represents kinetic state space for the quadruped robot. This representation intelligently and seamlessly integrates the four key quadruped robot proprioceptive state estimation components, marking the first time this has been carried out. The structure of the developed quadruped robot kinetic state-space model is shown in Figure 4.

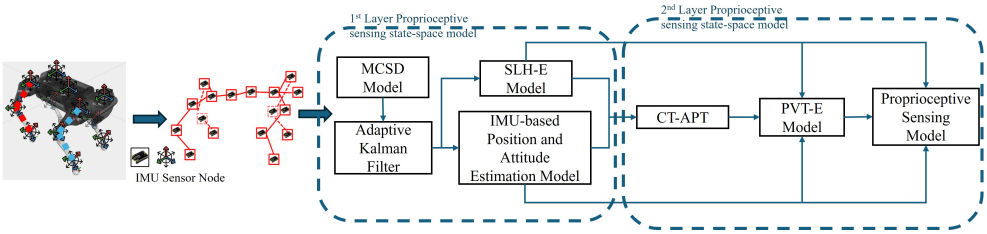


Figure 4. The developed quadruped robot proprioceptive state-space model.

Layer-1: Motion Pattern Model learning (High-level learning): The main goal of this layer is to process the raw data from multi-inertial sensors and generate the critical states that will be used for PVT estimation and quadruped robot proprioception in the next layer. Specifically, this layer includes four components, i.e.,

i Motion Classification and Step Detection (MCSD) Model: To better classify motion and detect gaits, this block develops a novel Time-Series Continuous-State Markov Model (TS-CSMM). By using time-series inertial sensing data, the motion classification and step detection accuracy can be significantly improved.

ii Adaptive Kalman Filter (AKF): The developed AKF effectively reduces the effects of environmental noise and sensor errors adaptively.

iii Supervise Gait Length and Heading Estimation (SLH-E) Model: A time-series probability mathematical model is generated that can better supervise the gait length and heading estimation online.

iv IMU-based Position and Attitude Estimation Model: Using the inertial sensing data along with inertial navigation computation, a position and attitude estimation model is built. It integrates with the SLHE model to enable 3D quadruped robot proprioceptive sensing.

Next, using the learned motion pattern model as well as estimated navigation parameters, the uncertainties from the environment and unreliable data are compensated, allowing more accurate quadruped robot kinetic and whole-body model.

Layer-2: Motion Pattern Guided Quadruped Robot Modeling (Low-level learning): This layer aims to generate a time-series statistical mathematical model for 3D quadruped robot proprioceptive state estimation under harsh environments, e.g., GPS-denied areas, etc. Specifically, there are three key components developed in this layer, i.e.,

i Continuous-Time Adaptive Particle Filter (CT-APT): To improve the accuracy of quadruped robot proprioceptive state-space modeling in real-time, the block CT-APT integrates a statistical approach with a particle filter.

ii Position, Velocity, Timing Estimation (PVT-E) Model: Integrating the SLH-E model with the IMU-based position and attitude estimation model from layer 1, a real-time PVT-E model is developed in Layer 2.

iii Quadruped Robot Proprioceptive State Model: Using the PVT-E model, the framework accurately computes the distance traveled as well as the elapsed time. A practical time-based objective moving model is generated in a 3D space, which is defined as the quadruped robot proprioceptive state model in this paper.

For proprioceptive modeling, vector p_i is constructed to represent values of 3D locations of joints relative to the body of the robot. And vector f is constructed to hold the ground reaction forces for each leg. This takes us to the goal of the presented development with a joint environment and quadruped robot state space as

$$x = [\vec{p}^T, \dot{\vec{p}}^T, \vec{\omega}^T, \dot{\vec{\Theta}}^T]^T \qquad u = f \tag{2}$$

2.2. Hybrid Dynamic Control Architecture

Quadruped robots operating in an unstructured and uneven terrain can be modeled as hybrid systems defined within a hybrid system framework to perform a quantitative analysis of gait scheduler or step planner. The framework proposed in this paper considers a quadruped robotic system as two coupled components, i.e., a floating base and four two-link legs. A step by step modeling is presented as follows: Numerous frameworks appear in literature for modeling such systems; these include [14,15]. According to the notations of the general framework, flow set (continuous-dynamics) is denoted by $C \in \mathbb{R}^n$, the jump set (discrete-dynamics) is denoted by $D \in \mathbb{R}^n$, the deferential equation is described by the flow map $F : \mathbb{R}^n \to \mathbb{R}^n(\dot{x} = F(x))$, and the difference equation is defined by a jump map $G : \mathbb{R}^n \to \mathbb{R}^n(x^+ = G(x))$. Then a hybrid dynamical system $\mathcal{H}$ can be expressed as

$$\mathcal{H} := \begin{cases} x \in C & \dot{x} = F(x) \quad (Continuous - Dynamics) \\ x \in D & x^+ = G(x) \quad (Discrete - Dynamics) \end{cases} \tag{3}$$

Collectively, the tuple (C, F, G, D) is called the data of $\mathcal{H}$ybrid system.

Next, a controller that finds a series of optimal reaction forces for one gait cycle of locomotion by using MPC with a floating base model is expressed as [7,9,11]

$$\begin{aligned} m\ddot{p} &= \sum_{i=1}^{n_c} f_i - c_g \\ \frac{d}{dt}(I\omega) &= \sum_{i=1}^{n_c} r_i \times f_i \end{aligned} \tag{4}$$

where p, f_i, and c_g represent the robot's vectored position, reaction force, and gravitational force in the world frame. m is mass, and n_c is the number of contacts. $I \in \mathbb{R}^{3\times3}$ is the rotational inertial matrix.

Using the Hybrid Systems Framework Equation (4) is rewritten as

Continuous-Dyanmics:

$$\begin{cases} m\ddot{p} = \sum_{i=1}^{n_c} f_i - c_g \\ d(I\omega) = \sum_{i=1}^{n_c} (r_i \times f_i)dt \end{cases} \tag{5}$$

Discrete-Dynamics:

$$\begin{cases} m\dot{p}^+ - m\dot{p}^- = \sum_{i=1}^{n_c} \delta f_i \\ I\omega^+ - I\omega^- = \sum_{i=1}^{n_c} (r_i \times \delta f_i) \end{cases} \tag{6}$$

with $\dot{p}^+, \dot{p}^-$ being the robot velocity and ω^+, ω^- being the robot angular velocity right before and after the discontinuous event (environmental impact).

The second controller is the body joint angle-torque controller, which takes in the reaction forces from the above equations and utilizes it to give joint torque, position, and velocity. The dynamic equation can be given in the similar form as [7,9,11]

Continuous-Dynamics:

$$A\begin{pmatrix} \ddot{q}_f \\ \ddot{q}_j \end{pmatrix} + b + g = \begin{pmatrix} 0_6 \\ \tau \end{pmatrix} + J_c^T f_r \tag{7}$$

Discrete-Dynamics:

$$A\begin{pmatrix} \dot{q}_f^+ \\ \dot{q}_j^+ \end{pmatrix} - A\begin{pmatrix} \dot{q}_f^- \\ \dot{q}_j^- \end{pmatrix} = J_c^T \delta f_r \tag{8}$$

where A, b, τ, f_r, J_c are the generalized mass matrix, Coriolis force, gravitation force, joint torque, ground reaction force, and contact Jacobian, respectively. $\ddot{q} \in \mathbb{R}^6$ is the acceleration of the floating base, and $\ddot{q}_j \in \mathbb{R}^{n_j}$ is the vector of joint accelerations where n_j is the number of joints. $\dot{q}^+\dot{q}^-$ are the velocity of float base and joints before and after discontinuous events (environmental impact).

2.3. Proprioceptive Sensing Assisted Model Predictive Control

Model predictive control used in this framework produces a series of ground reaction forces that ensure the center of mass (CoM) follows the given trajectory. A predefined gait scheme with relaxed time bounding is fed, which makes this formulation a convex optimization problem. This makes it fast and easy to find the global minimum, unlike other nonlinear optimization formulations.

The floating base model, when approximated as a simple lumped mass, is prone to linearization errors due to the presence of the cross product term in Equation (4). To fit this into convex MPC formulation, the following three assumptions [8] are necessary: The first is that the roll and pitch angles are minor enough to be approximated from previous and next iteration steps. The second assumption is more related to the relaxed CoM trajectory

tracking, which monitors the states the quadruped robot ends up in. If they are close enough to the commanded trajectory, they are qualified as executed. This assumption is further equated to the models saved inside the memory of robot. This is further presented in the event-triggered section. The last assumption is that the pitch and roll velocities and off-diagonal terms of inertia tensors are small enough to be ignored. With these assumptions, the following equations are listed:

$$\begin{aligned} \dot{\Omega} &\approx R_z(\psi)\omega, \\ I_G &\approx R_z(\psi) I_B R_Z(\psi)^T \end{aligned} \tag{9}$$

$$\frac{d}{dt}(I\omega) = I\dot{\omega} + \omega \times (I\omega) \approx I\dot{\omega} \tag{10}$$

where $\dot{\Omega} = [\dot{\phi} \quad \dot{\theta} \quad \dot{\psi}]^T$ is the Euler angle representation for the roll, pitch, and yaw for the robot. $R_z(\psi)$ is the rotational matrix. I_G, I_B are the inertial tensor forces in global and body frame, respectively.

With these assumptions, the discrete dynamics of the system is expressed as

$$x(k+1) = A_k x(k) + B_k f(k) + g, \tag{11}$$

where,

$$\begin{aligned} x &= [\vec{p}^T, \dot{\vec{p}}^T, \vec{\omega}^T, \dot{\vec{\Theta}}^T]^T \\ f &= [f_1 \cdots f_n]^T \\ g &= [0_{1\times3} \quad 0_{1\times3} \quad 0_{1\times3} \quad g^T]^T \end{aligned} \tag{12}$$

$$A = \begin{bmatrix} 1_{3\times3} & 0_{3\times3} & R_{z(\psi_k)}\Delta t & 0_{3\times3} \\ 0_{3\times3} & 1_{3\times3} & 0_{3\times3} & 1_{3\times3}\Delta t \\ 0_{3\times3} & 0_{3\times3} & 1_{3\times3} & 0_{3\times3} \\ 0_{3\times3} & 0_{3\times3} & 0_{3\times3} & 1_{3\times3} \end{bmatrix} \tag{13}$$

$$B = \begin{bmatrix} 0_{3\times3} & \cdots & 0_{3\times3} \\ 0_{3\times3} & \cdots & 0_{3\times3} \\ I_G^{-1}[r_1]_\times \Delta t & \cdots & I_G^{-1}[r_n]_\times \Delta t \\ 1_{3\times3}\Delta t/m & \cdots & 1_{3\times3}\Delta t/m \end{bmatrix} \tag{14}$$

A general quadratic programming formulation is constructed using the above equations, which minimizes the state cost function as

$$\min_{x,f} \sum_{k=0}^{m} \|x(k+1) - x^{ref}(k+1)\|_Q + \|f(k)\|_R \tag{15}$$

subject to dynamics and initial constraints shown in Algorithm 1.

Algorithm 1 Trajectory Optimization for Legged Robots

1: **System Input (User Defined):**
Initial state of the system: $r(0), \theta(0), p_i(0), f_i(0)$
Desired final state: $r(T), \theta(T)$
Total time horizon: T
Number of steps for each foot: $n_{s,i}$

2: **Initialize Decision Variables:**
CoM trajectory: $r(t) \in \mathbb{R}^3$
Orientation: $\theta(t) \in \mathbb{R}^3$
Foot positions: $p_i(t) \in \mathbb{R}^3$
Contact forces: $f_i(t) \in \mathbb{R}^3$
Phase duration for each foot: $\Delta T_{i,j} \in \mathbb{R}$

3: **Decide the Objective:**
Minimize the total energy or effort required for the motion:

$$\min \sum_{i=1}^{n} \int_0^T \|f_i(t)\|^2 dt$$

4: **Dynamics Constraints:**
Enforce dynamics constraints that relate CoM acceleration to contact forces given by equations in (9) and (10) above

$$m\ddot{r}(t) = \sum_{i=1}^{n} f_i(t) - mg$$

$$I\dot{\omega}(t) + \omega(t) \times I\omega(t) = \sum_{i=1}^{n} f_i(t) \times (r(t) - p_i(t))$$

5: **Kinematic Constraints:**
Ensure that each foot stays within its workspace:

$$p_i(t) \in R_i(r(t), \theta(t))$$

6: **Contact and Force Constraints:**
Forces are only applied when feet are in contact:

$$f_i(t) = 0 \qquad \text{if foot } i \text{ is not in contact}$$

Non-slipping condition (Coulomb's law):

$$\sqrt{f_{t1}^2 + f_{t2}^2} < \mu f_n$$

where f_n is the normal force, and μ is the friction coefficient.

7: **Gait Discovery and Phase Duration:**
Use phase-based parameterization to optimize swing and stance duration $\Delta T_{i,j}$:

$$p_i(t) = \begin{cases} \text{cubic polynomial} & \text{during swing phase,} \\ \text{constant} & \text{during stance phase.} \end{cases}$$

8: **Terrain Constraints:**
Ensure that foot height matches terrain height during stance phases:

$$p_z(t) = h_{\text{terrain}}(p_{xy}(t)) \quad \text{if foot } i \text{ is in contact}$$

9: **Solve the QP using formulation in Equation (15):**
Optimize $r(t), \theta(t), p_i(t), f_i(t), \Delta T_{i,j}$

10: **Output:**
Optimized trajectories for: CoM position $r(t)$, Orientation $\theta(t)$, Foot positions $p_i(t)$, Contact forces $f_i(t)$

2.4. Motion Patterning and Terrain Adaptation

While proprioception is essential for state estimation and precise modeling of legged robots, exteroceptive sensors (IMU) provide crucial information about the robot's surroundings, facilitating tasks such as navigation, obstacle avoidance, terrain analysis, and object manipulation. This subsection presents the significance of proprioception and exteroception in modeling and control design. Recognizing the high computational cost of processing high-dimensional data in real-time, a low-dimension embedding of the sensory data, as shown in Figure 4, is learned, allowing for more efficient state estimation and environmental dynamics modeling. The low-dimensional data embedding is integrated into the proposed framework for state estimation and joint modeling. Figure 5 shows the process by which the multi-modal architecture is trained. The goal is to train N terrain models to comply with the equation $F_S = [f_1(x,u) f_2(x,u) \dots F_N(x,u)]$, which have different action spaces, and one model is triggered at one point in time. The triggering is completed by a switching mechanism that compares the real-time quadruped robot movement with estimated movement from an individual model that is trained as shown in Figure 6.

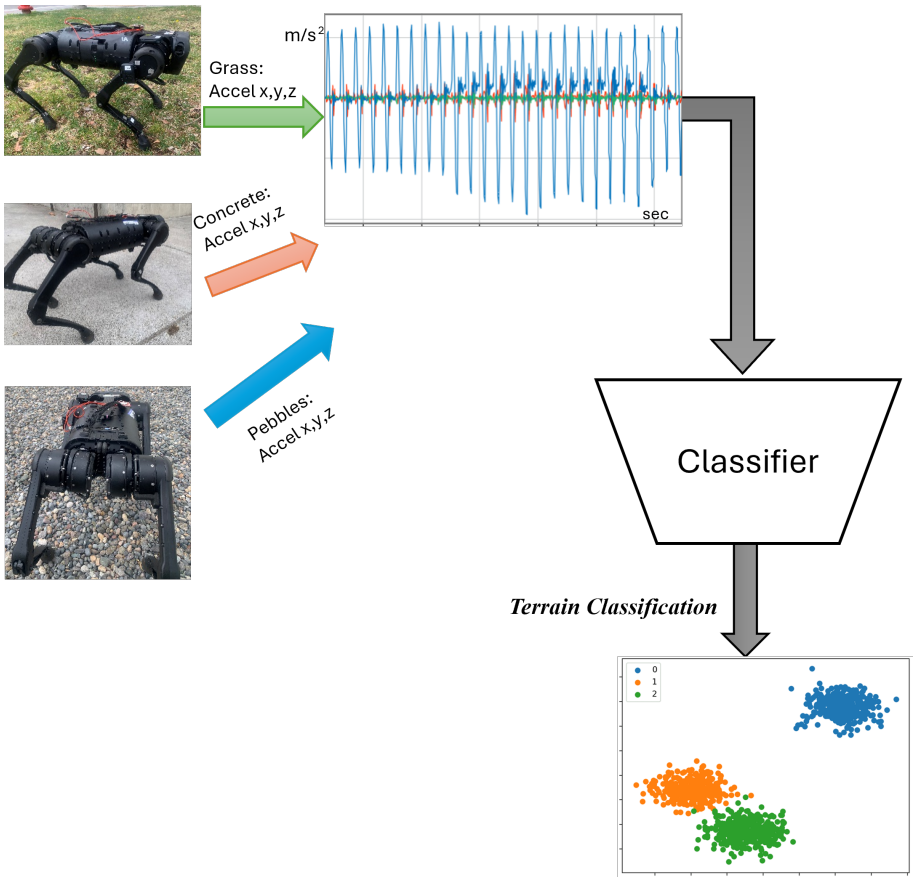


Figure 5. Situation-driven multi-modal terrain classification architecture. The images on the left show the quadruped robot navigating four different terrains: grass (green arrow), concrete (orange), and gravel (blue). Each arrow points to the corresponding segment of multi-IMU raw signal data (top center plot), where colors (same as arrows) indicate different terrain. These signals are processed by a classifier (center) trained to distinguish terrain types. The scatter plots at the bottom right show terrain classification results via PCA projection, with colored dots representing clustered IMU samples. Group 0, Group 1, and Group 2 correspond to pebbles, concrete, and grass resp.

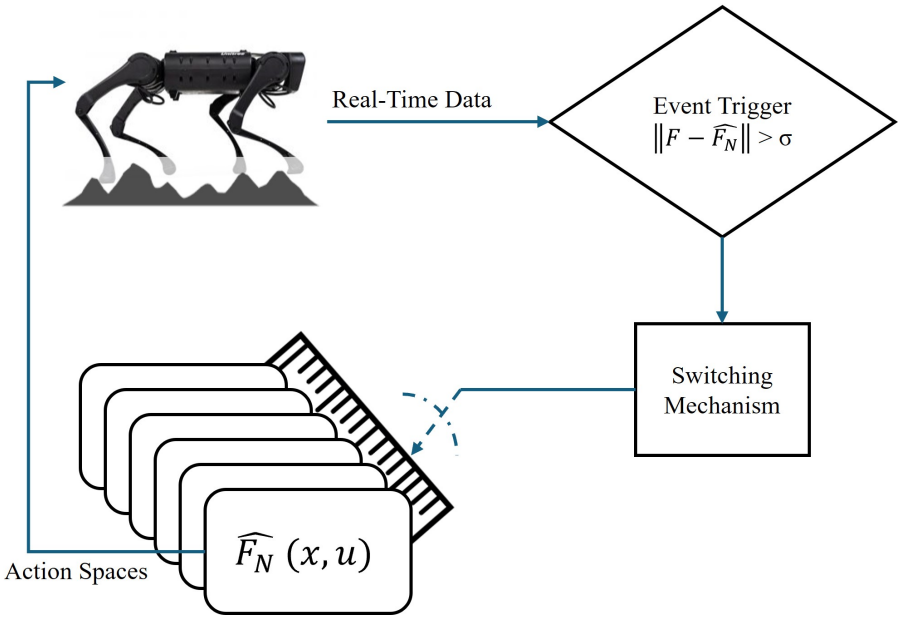


Figure 6. The structure of discontinuous event-triggered hybrid model.

2.4.1. Deep State Space for Continuous Motion Patterning

This part shows the development of the mechanism that estimates the practical time-based continuous motion pattern of the quadruped robot. The key idea here is to extend the traditional spatial motion pattern state space $f_{s,i}(*)$ into a novel spatio-temporal motion pattern state space $f_{ST}(*,t)$. According to [34,35] existing quadruped robot modeling research, the spatial motion pattern state space effectively transforms the collected IMU data, i.e., acceleration and angular rates (a, w), to motion parameters, i.e., $\Delta l, \Delta\psi$, under and identified motion pattern as:

$$(\Delta l, \Delta\psi|\theta_i) = f_{s,i}(a, w, v(0), g(0)|\theta_i), \qquad i = 1, 2, \ldots, N \tag{16}$$

Here, $v(0)$ and $g(0)$ represents the initial velocity and gravity, respectively; θ_i represents the classified motion patterns under different terrains (e.g., walking on ice, running on grass, slow walking on rocky road, etc.), and $f_{s,i}(*)$ represents the i-th type of spatial motion pattern. It is important to note that quadruped robots might change their movement patterns continuously. Therefore, roughly switching among finite classified motion patterns might significantly degrade the parameter $\Delta l, \Delta\psi$ estimation performance and seriously affect the accuracy of quadruped robot kinetic and whole-body modeling. To handle this intractable challenge, a novel spatial-temporal deep state space for representing continuously patterning trainees' motion in real-time is given as follows:

$$(\Delta l(t), \Delta\psi(t)) = f_{ST}(a(t), w(t), v(0), g(0), t) = \sum_{i=1}^{N} W_i f_{S,i}(a, w, v(0), g(0)|\theta_i) \tag{17}$$

Compared with the finite motion pattern set, $F_s(*) = f_{S,1}(*), \ldots, f_{S,N}(*)$ that has been generated through spatial IMU data, our developed unique deep state-space-based continuous motion pattern $F_{ST}(*,t)$ is a time-based function that has been trained via spatio-temporal IMU data.

2.4.2. Online Motion Pattern Guided Quadruped Robot Model

The main tenet of the developed motion pattern guided model hinges on the following two key ideas:

1. Building a hierarchical deep learning architecture that processes the collected IMU data in different resolution levels to upgrade the accuracy level of quadruped robot modeling
2. Operating a two-layer deep learning algorithm in parallel that reduces the computational complexity and promotes its real-time efficiency.

Specifically, the proposed hierarchical deep learning is designed as

Layer 1: Motion Pattern Model learning (High-level learning)

A deep state-space continuous motion pattern model is trained through a novel 2-stage spatio-temporal NN:

$$(\Delta\hat{l}(t), \Delta\hat{\psi}(t)) = f_{ST}(a(t), w(t), v(0), g(0), t) \tag{18}$$

Next, using the learned motion pattern model as well as estimated navigation parameters, the uncertainties from the environment and unreliable data are compensated, allowing a more accurate quadruped robot kinetic and whole-body model.

Layer 2: Motion Pattern Guided Quadruped Robot Modeling (Low-level learning)

To derive real-time navigation as well as deliver an accurate quadruped robot model, the starting location $L = (L_0^x, L_0^y)$ and heading ψ_0 , and the Cartesian projection can be represented as

$$\begin{cases} L_t^x = L_0^x + \Delta l * cos(\psi_0 + \Delta\psi) + g_x(\Delta l, \Delta\psi, t) \\ L_t^y = L_0^y + \Delta l * sin(\psi_0 + \Delta\psi) + g_y(\Delta l, \Delta\psi, t) \end{cases} \tag{19}$$

with $g_x(*), g_y(*)$ being the nonlinear uncertainties from the environment. It is important to note that $\Delta l(t), \Delta\psi(t)$ can be obtained from the learned motion pattern model in Layer-1. However, even if one is able to generate a high-accuracy motion pattern model, the uncertainties $g_x(*), g_y(*)$ still cannot be fully compensated and will degrade the quadruped robot modeling and navigation performance. Therefore, a deep neural network-based uncertainty estimator is developed. It is used along with the learned motion pattern model to conduct high-accuracy and real-time quadruped robot modeling and navigation.

$$\begin{cases} \hat{L}_t^x = L_0^x + \Delta\hat{l} * cos(\psi_0 + \Delta\hat{\psi}(t)) + \hat{g}_x(\Delta\hat{l}, \Delta\hat{\psi}, t) \\ \hat{L}_t^y = L_0^y + \Delta\hat{l} * sin(\psi_0 + \Delta\hat{\psi}(t)) + \hat{g}_y(\Delta\hat{l}, \Delta\hat{\psi}, t) \end{cases} \tag{20}$$

with the DNN-based uncertainty estimation being designed as

$$\hat{g}_x\left(\Delta\hat{l},\ \Delta\hat{\psi}, t\right) = \hat{W}_{gx}^T(t)\varphi_x\left(\Delta\hat{l}, \Delta\hat{\psi}, t\right);\ \hat{g}_y\left(\Delta\hat{l}, \Delta\hat{\psi}, t\right) = \hat{W}_{gy}^T(t)\varphi_y(\Delta\hat{l}, \Delta\hat{\psi}, t) \tag{21}$$

and DNN weights are updated as

$$\dot{\hat{W}}_{gx}(t) = -\alpha_{gx}\frac{\varphi(\Delta\hat{l}, \Delta\hat{\psi}, t)e_{gx}^T}{\|\varphi_x\left(\Delta\hat{l}, \Delta\hat{\psi}, t\right)\|^2 + 1}; \dot{\hat{W}}_{gy}(t) = -\alpha_{gy}\frac{\varphi(\Delta\hat{l}, \Delta\hat{\psi}, t)e_{gy}^T}{\|\varphi_y\left(\Delta\hat{l}, \Delta\hat{\psi}, t\right)\|^2 + 1} \tag{22}$$

$$e_{gx}(t) = \sqrt{\int_{t-\Delta t}^{t} \|L_\tau^x - \hat{L}_\tau^x\|^2 d\tau} \quad ; e_{gy}(t) = \sqrt{\int_{t-\Delta t}^{t} \|L_\tau^y - \hat{L}_\tau^y\|^2 d\tau} \tag{23}$$

Through hierarchical deep learning development, there is an improvement of the situation-aware quadruped robot kinetic and whole-body modeling and navigation accuracy and time efficiency through parallel computing shows the real deployment structure.

For the experimental results in a real-world setup, three different terrain classifications were used, and an unknown, highly uneven terrain was tested. With the help of this new

mechanism, the accuracy of the model in real-world applications is increased. The final form of the equation with terrain uncertainties is written as

$$F(x, f, \theta) = \sigma[F_1(x, f, \theta_1), F_2(x, f, \theta_2), \ldots, F_N(x, f, \theta_N)] \tag{24}$$

with $F_i(x, f, \theta_i) i = 1, 2, \ldots, N$ being the situation-aware dynamics model space basis set and N being the dimension of the constructed situation space. $F(x, f, \theta)$ is the real-time situation-aware robot dynamic model that is unknown due to the uncertainties for practical situations and terrains. Through effectively incorporating the uncertainties into the modeling, Equation (24) can now solve the model predictive control with better accuracy.

3. Results

3.1. Experimental Setup and Results for Multi-IMU Enhanced State Estimation

The Table 1 shown below details the specifications of IMUs used for Figure 3.

Table 1. IMU Sensor Specifications and Placement.

Sensor ID	Location	Orientation	Model	Rate (Hz)
IMU-1	Torso (CM)	Fwd, Left, Up	x-IMU3	200
IMU-2	Torso-to-Shoulder	Fwd, Left, Up	x-IMU3	200
IMU-3	Upper Shoulder	Fwd, Left, Up	x-IMU3	200
IMU-4	Upper heel	Fwd, Left, Up	x-IMU3	200

The initial state variables for PVT estimation architecture (Figure 4) are shown in Table 2. The CoM position is estimated with reference to the world frame. And the initialization is always (0,0,0). Note the standing position CoM is considered $z = 0$. If the robot has to bend down, then the z position is in negative.

Table 2. Initial Values of State Variables for Control and Estimation.

State Variable	Symbol	Initial Value	Description
Position (CoM)	x, y, z	(0, 0, 0) [m]	Body CoM pos
Orientation (Euler angles)	ϕ, θ, ψ	(0, 0, 0) [rad]	Euler upright angles
Linear velocity	$\dot{x}, \dot{y}, \dot{z}$	(0, 0, 0) [m/s]	CoM linear velocity
Angular velocity	$\omega_x, \omega_y, \omega_z$	(0, 0, 0) [rad/s]	Body angular rate
Joint angles	$\mathbf{q}$	Nominal stance	Leg configuration
Joint velocities	$\dot{\mathbf{q}}$	0 [rad/s]	Initial joint angles
Terrain state	terrain_class	Flat/neutral class	Init Terrain est.
Filter covariance matrix	P_0	$0.01 \cdot I$	Initial uncertainty

The initial values from Table 2 are fed to the Control System (MPC/PD) initialization for planning and foot placement execution as given in the following equation:

$$r = \begin{bmatrix} x & y & z & \dot{x} & \dot{y} & \dot{z} & \phi & \theta & \psi & \dot{\phi} & \dot{\theta} & \dot{\psi} & \omega \end{bmatrix}$$

A series of simulations were performed in robot-operating-system-based environment [37] for different terrain testings. Figure 7 shows a still capture of the simulation. The robot model is provided in unified robot description format (URDF) by the makers of unitree legged robot. In simulation environments, terrain physicality often differs significantly from real-world conditions, primarily due to the lack of debris variability. Unlike actual terrains, which include diverse and unpredictable elements such as rocks, sticks, loose soil, and other irregular obstacles, simulated terrains tend to have uniform or simplified surface models. This absence of varied debris leads to less realistic interaction

dynamics between the robot and the terrain, potentially causing discrepancies in locomotion performance, traction, and stability when transitioning from simulation to real-world deployment. In addition, there is no absolute metric comparison between simulation and real-world experiments; instead, the analysis focuses solely on relative behavioral trends observed across both domains. This approach acknowledges inherent differences in environmental complexity and measurement conditions, emphasizing comparative patterns rather than exact numerical equivalence.

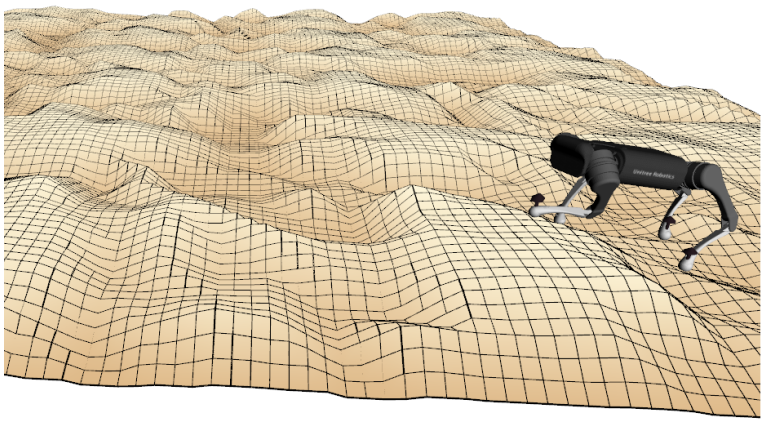


Figure 7. Terrain Classification Testing in Simulation.

A comparison between the simulation and real-world test conditions is shown in Table 3. Further a concise summary of training and testing parameters are shown in Table 4.

Table 3. Comparison Between Simulation and Real-World Test Conditions.

Feature	Simulation	Real-World	Match
Terrain Types	Modeled gravel, sand, concrete	Natural grass, gravel, sand, concrete	Partial
Commanded Speed	0.5–1.2 m/s	0.5–1.2 m/s	Yes
Payload Mass	15 kg + sensors (simulated)	15.2 kg with onboard sensors	Yes
IMU Placement	URDF-based virtual placement	Real placement as per URDF	Yes
Sensor Data Fidelity	Clean synthetic signals	Noisy, terrain-influenced IMU data	No
Environmental Effects	Controlled, friction-tuned mesh	Unpredictable terrain (debris, wetness)	No

Table 4. Summary of Dataset for Training and Testing of Deep Learning Models.

Model	Source	Time (min)	Samples	Window	Labels	Split	Augmentation
Terrain Classifier	Real robot IMU	72	∼864,000	500 (2.5 s)	Supervised	70/15/15	Yes
PVT Estimator	IMU + MoCap/GPS	33	∼1.4 M	1000	Supervised	70/15/15	No

For the developed real robot multi-IMU state estimation experiments, MoCap Optitrack has been used to provide accurate movement of quadruped robot, i.e., ground truth. Results are shown in Figure 8 and Table 5 with root mean square errors use for evaluating position estimation efficiency.

Our experiments demonstrated that the developed proprioception mechanism is sufficient for state estimation, and the designed approach without using data from vision systems can still achieve competitive accuracy compared to traditional state estimation techniques such as extended Kalman filter (EKF) with the mocap measurements and so on.

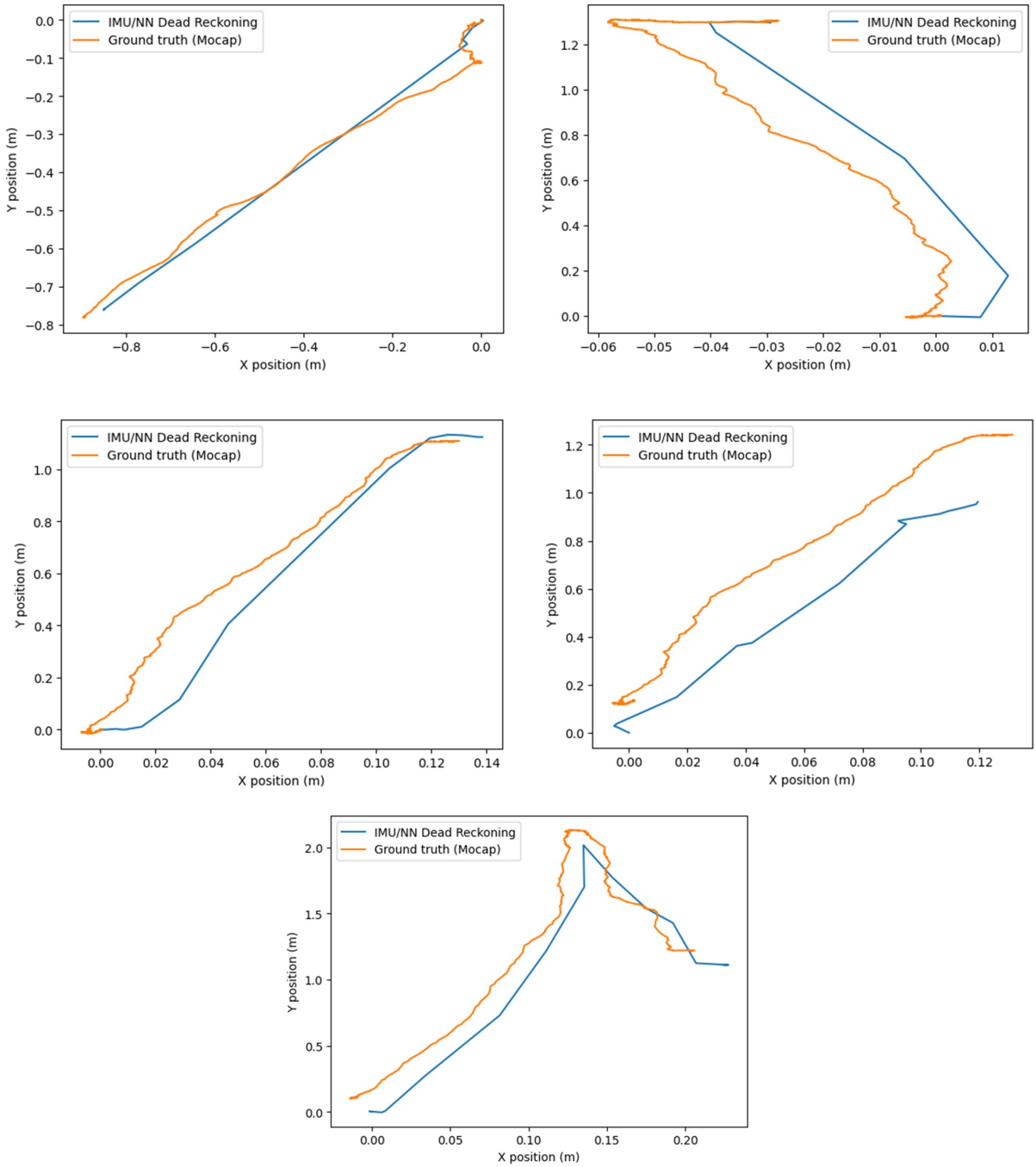


Figure 8. Indoors Random Walk experiments with MoCap.

Table 5. Indoors Experiment for Position Estimation.

Experiment #	% Error
1	2.21
2	4.63
3	3.78
4	4.79
5	4.07

3.2. Experimental Results of Terrain Classification Enhanced Switching Controller

A few experiments have been conducted under challenging terrains and further evaluated the effectiveness of the developed technique by comparing it with other existing techniques. The results are shown as follows:

1. Using traditional model predictive control [8] as well as learning-based control (Rapid Motor Adaption) [38] under challenging terrain, e.g., rocks are bigger than the legged robot's foot diameter, flat surface with uncertain wholes bigger than the legged robot foot diameter.
   Video Link 1: (Mixed Terrain 1) https://nevada.box.com/s/bmvm6hnjrvit793xwlzk7cu7z2poam0i (accessed on 27 June 2025)
   Video Link 2: (Mixed Terrain 2) https://nevada.box.com/s/56ep8272uyojt2icrzaptfb1m2sopbba (accessed on 27 June 2025)
2. Using our developed dual learning enhanced new generation terrain-proprioceptive hierarchical hybrid control for a quadruped robot in the presence of challenging terrain, e.g., rocks are bigger than legged robot's foot diameter, flat surface with uncertain wholes bigger than legged robot's foot diameter.
   Video Link 3: (Mixed Terrain 1—Our Methodology) https://nevada.box.com/s/c5y4rrdgu7zjj7mc8jsbqqjfzhspvtfc (accessed on 27 June 2025)
   Video Link 4: (Mixed Terrain 2—Our Methodology) https://nevada.box.com/s/45e0rs7djqdyhcycnurl37jiwxqes371 (accessed on 27 June 2025)

4. Discussion

Robot Platform and Hardware Setup

The proposed control framework was implemented and tested on a pre-built quadruped robot Unitree-A1 platform with a modular architecture. Each leg is equipped with three actuated degrees of freedom (hip, thigh, and knee), driven by high-torque brushless DC motors with integrated encoders for joint feedback. The robot's body is constructed from lightweight iron ore and carbon fiber composites, supporting a total payload of approximately 15 kg. For proprioceptive sensing, we integrated four low-cost MEMS-based inertial measurement units (IMUs) (x-IMU-3), strategically placed on the robot's torso and one leg segment (Figure 3). These IMUs provide real-time 6-DoF inertial data (acceleration and angular velocity) at 200 Hz. The sensors are connected to a central offboard computer (NVIDIA Jetson Xavier NX) wirelessly, which runs the hybrid dynamic modeling, sensor fusion, terrain classification, and switching MPC control loop in real time. All control algorithms were implemented in C++ and Python, with optimization routines solved using the OSQP QP solver. The full control pipeline runs at 120 Hz, while the terrain classifier operates asynchronously at 10 Hz. The system was tested indoors and outdoors over multiple terrain types, including flat concrete, grass, loose gravel, and sand, without relying on exteroceptive sensing.

Results

While there are a lot of highly dynamic robotic control implementations in the literature, most of them did not work well when the terrain was excessively uneven, as shown in the experiments (Link 1 and 2). In contrast to biological animals, which have a complex and versatile nervous system that gives instantaneous feedback from the ground reaction forces, which are almost independent of their brain's reaction signals, these quadruped robotic systems lack this instantaneous feedback system. The goal was to replicate this nervous system parallel to its skeleton and independent of the model predictive control's reaction force calculations. We set out with this research of replicating a "nervous system" of biological animals with the use of a network of IMUs, expecting to have an informed knowledge of the ground reaction forces. We were right in expecting the information would

help in determining the right amount of torque required by each actuator; however, we were wrong in assuming the same torque calculations can be applied to all kinds of terrain. For example, we incorrectly assumed that we could use the same torque calculations for sandy surfaces and loose gravel with miniature rocks. The torque calculations for loose gravel would overshoot, while for sandy surfaces it was insufficient. Interestingly, for sandy surfaces and wet/muddy surfaces, the control calculations were relatively the same for a stable gait as opposed to the torque calculations for sandy surfaces and loose gravel. Hence, we decided to put an extra layer of terrain classification as an added knowledge base aiding torque calculations for a stable gait control across different terrain parameters. What surprised us was even though we had a small set of discrete terrain classifications, we were still able to navigate most of the previously unsuccessful experiments. From here, our next goal is to artificially generate intermediate imaginary terrain classifications between similar terrains and enlarge the model knowledge base for real-world applications without requiring prior training.

Figures 9 and 10 show the indoor experiment with position, velocity, acceleration, and ground reaction forces experienced by the quadruped robot with the commanded velocities as 0.5 m/s and trotting and pacing gaits as 1.0 m/s. This shows that state estimation and velocity command tracking are efficient and effective for indoor environments. Next, for the outdoor experiments, videos given in links 1 and 2 show the quadruped robot cannot maintain stability while using traditional reinforcement learning-enhanced controllers. The inability of the quadruped robot to determine the underlying uncertainty is apparent. However, when the quadruped robot is equipped with the developed situation-aware bi-level hybrid switched controller, its performance, including stability and resiliency, has been significantly improved even under uncertain outdoor environments. To benchmark the effectiveness of our proprioceptive switching MPC framework, we compared it against traditional MPC and reinforcement learning-based controllers across a variety of terrain conditions (e.g., rocky, sandy, and mixed debris). As shown in Table 6, our method achieves a significantly lower position RMSE (0.5 m vs. 0.84 m for standard MPC), better velocity tracking, and a drastic reduction in instability events. Moreover, our framework maintains a fast average control cycle of 20 ms, enabling real-time deployment. These results confirm that integrating proprioceptive sensing with situation-aware switching leads to improved stability, responsiveness, and robustness, especially under uncertain terrain conditions.

Strengths, Limitations, and Generalizability

In summary, some of the key strengths of our methodology include terrain adaptability, sensor robustness, scalability, and modularity and real-time performance. In addition to these improvements, the authors also want to highlight some of the limitations still to be addressed in this field of research and hope to work on it in the future. These include: sensor drift, model dependency, and terrain class limitations. The low-cost IMUs, while fast and power-efficient, suffer from long-term drift. This "long-term" parameter varies with specific locomotion patterns, and periodic correction needs to be addressed in more depth. In terms of model dependency, our system, as with any other state-of-the-art implementation, does require prior knowledge of reasonably accurate dynamics models in the learning/data collection phase. Mismatches in these labels will degrade the performance. For terrain classification, while the authors had spent quite a lot of time gathering raw data for different terrains, intermediate or unseen terrain traversal was the main cause of reported instability in Table 6. Our future work will address this limitation via online learning methods.

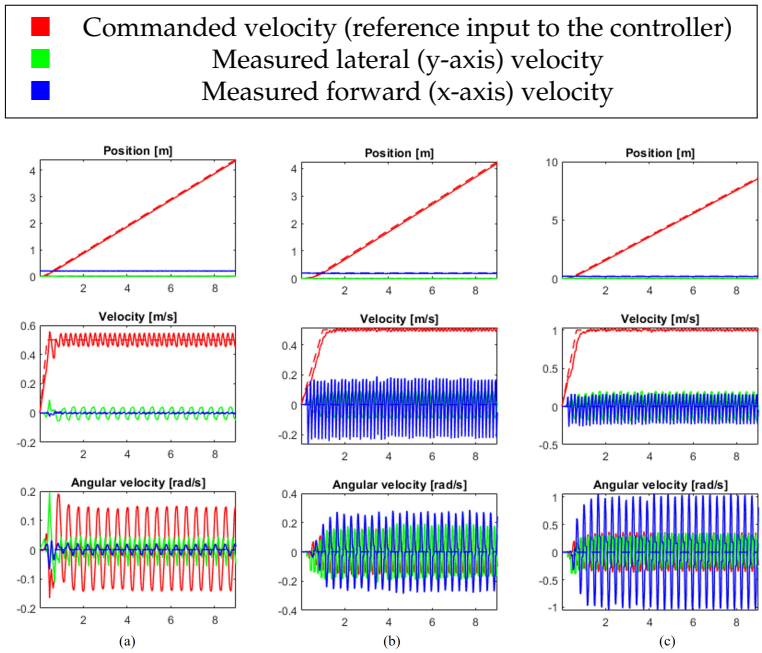


Figure 9. Indoors Experiment. (**a**) Commanded Velocity 0.5 m/s | Gait Trot (**b**) Commanded Velocity 0.5 m/s | Gait Run (**c**) Commanded Velocity 1.0 m/s | Gait Pacing.

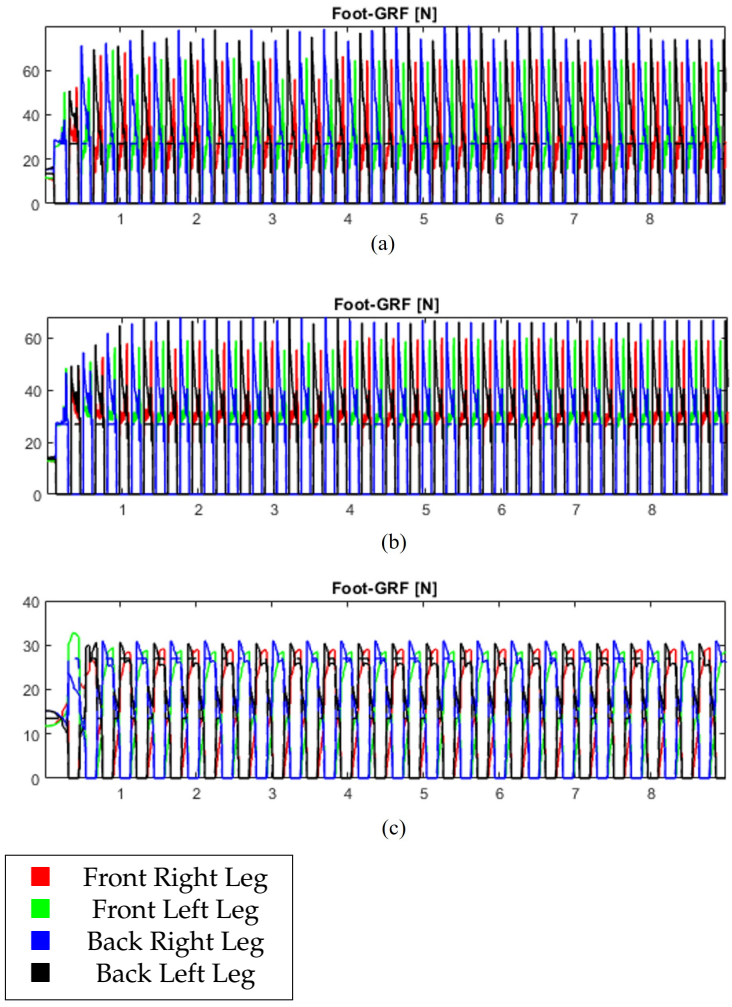


Figure 10. Indoors Experiment Ground Reaction Forces. (**a**) Commanded Velocity 0.5 m/s | Gait Trot (**b**) Commanded Velocity 0.5 m/s | Gait Run (**c**) Commanded Velocity 1.0 m/s | Gait Pacing.

Table 6. Key Performance Indices comparing our Methodology with Traditional MPC- and RL-based methods.

KPI	MPC *	RL Control *	Proposed Method
Position RMSE (m)	0.84	0.56	0.5
Velocity Tracking Error (m/s)	1.2	0.76	0.6
Instability Events per Trial	7.8	1.8	1.2
Terrain Recovery Success Rate (%)	65	82	88

* The implementations of traditional and RL-Based Control of Unitree A1 are closed source and the tuning parameters used in the respective papers are unknown. All experiments with the above mentioned methodologies were conducted with the same tuning parameters as our proposed methodology.

For researchers looking to adapt this methodology into their custom quadruped robot, the following generalization points need to be kept in mind:

1. Replacing the robot-specific dynamic model components with those tailored to the new platform (e.g., different limb morphology).
2. Adjusting the IMU sensor placements and retraining the terrain classification module using the new robot's data.
3. Tuning MPC parameters and constraints to match the actuation capabilities and gait styles of the target robot.

This modular structure—combining proprioception, hybrid modeling, and switching control—makes the framework applicable to other legged platforms such as hexapods or humanoids, provided similar proprioceptive sensing and control primitives are available.

Contextualizing with Embodied Adaptive Control Approaches

Recent developments in biologically inspired control architectures have demonstrated the potential of embodied adaptive systems, which mimic neuromotor control seen in biological organisms. These systems often leverage spiking neural models, neuromechanical feedback, and dynamic learning to produce adaptive and robust locomotion. In particular, recent work on embodied motor neuron architectures has shown promise for integrating sensory feedback and control within the same computational units, offering low-latency, energy-efficient control mechanisms for dynamic and unstructured environments. Similarly, nullcline-based control strategies for piecewise-linear (PWL) oscillators provide a principled method to generate stable limit cycles and entrained gaits, with applications in rhythmic locomotion over uneven terrains. While our proposed framework does not explicitly adopt these bio-inspired mechanisms, it shares a number of design principles with embodied approaches:

- It emphasizes distributed feedback through proprioceptive IMUs (analogous to biological sensory neurons).
- It uses hierarchical and switching control, akin to how biological systems coordinate reflexive and voluntary motion layers.
- It enables real-time adaptation to terrain perturbations, which is central to embodied locomotion control.

Future work can explore hybridizing the proposed model with embodied neuron-inspired architectures—for example, by replacing the switching logic with nullcline-based oscillators or integrating adaptive spiking neuron-based feedback for low-level torque modulation. This direction holds promise for further reducing computational cost while increasing biological plausibility and terrain generalization.

5. Conclusions

This paper presented a novel proprioception-enhanced control framework for quadruped robots operating in uncertain and unstructured terrains. By combining a situation-aware bi-level hybrid dynamic model with a three-layer switching model predictive control (MPC) strategy, the proposed system achieves high adaptability, low latency, and improved robustness—even in the absence of exteroceptive sensing like cameras or LiDAR.

Key Results from experimental evaluations show:

- A significant reduction in position estimation error (RMSE decreased by over 40% compared to traditional MPC).
- A 88% terrain recovery success rate on uneven surfaces where baseline controllers failed.
- Real-time operation with average control cycle latency under 20 ms.
- Improved stability and gait adaptability across various terrains, including gravel, grass, and sand.

Future Work

Building on this foundation, future research will focus on:

- Expanding the terrain classification space with self-supervised learning to include unknown or intermediate terrains.
- Integrating embodied motor neuron architectures and nullcline-based control for more biologically plausible and energy-efficient behavior.
- Transferring this framework to other legged platforms (e.g., hexapods or bipedal robots) and testing it under real-world field conditions.
- Exploring tighter integration between the perception and control layers using spatiotemporal deep neural networks for gait prediction and control fusion.

Author Contributions: S.L. and H.X. are with the Electrical and Biomedical Engineering at the University of Nevada, Reno. Y.B., P.C., D.S. and G.C. are with Intelligent Fusion Technology Inc., MD., USA. Conceptualization, S.L. and H.X.; methodology, H.X.; software, S.L.; validation, S.L, Y.B. and P.C.; formal analysis, S.L.; investigation, H.X. and S.L.; resources, S.L.; data curation, S.L.; writing—original draft preparation, S.L. and H.X.; writing—review and editing, H.X.; visualization, S.L.; supervision, H.X.; project administration, D.S.; funding acquisition, G.C. All authors have read and agreed to the published version of the manuscript.

Funding: This material is partially based upon work supported by the U.S. Army Small Business Innovation Research Program Office and the Army Research Office under Contract No. W911NF-24-P-0008. The authors thank Dr. Dean Culver from Army Research Office for his technical guidance and suggestions during the effort. Any opinions, findings and conclusions or recommendations expressed in this material are those of the authors and do not necessarily reflect the views of the U.S. Army Small Business Innovation Research Program Office and the Army Research Office.

Data Availability Statement: The datasets presented in this article are not readily available because the datasets are part of an ongoing study/proposal. Requests to access the datasets should be directed to the author Dr. Hao Xu.

Conflicts of Interest: Authors Yajie Bao, Peng Cheng, Dan Shen and Genshe Chen were employed by the company Intelligent Fusion Technology, Inc. Author Sanket Lokhande was pursuing his PhD. at the University of Nevada, Reno at the time of writing this paper. The remaining authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

References

1. Dudzik, T.; Chignoli, M.; Bledt, G.; Lim, B.; Miller, A.; Kim, D.; Kim, S. Robust autonomous navigation of a small-scale quadruped robot in real-world environments. In Proceedings of the 2020 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Las Vegas, NV, USA, 25–29 October **2020**; pp. 3664–3671.
2. Joseph, T.; Shaikh, A.; Sarode, M.; Rao, Y.S. Quadruped Robots: Gait Analysis and Control. In Proceedings of the 2020 IEEE 17th India Council International Conference (INDICON), New Delhi, India, 10–13 December 2020; pp. 1–6.
3. Medeiros, V.S.; Jelavic, E.; Bjelonic, M.; Siegwart, R.; Meggiolaro, M.A.; Hutter, M. Trajectory optimization for wheeled-legged quadrupedal robots driving in challenging terrain. *IEEE Robot. Autom. Lett.* **2020**, *5*, 4172–4179. [CrossRef]
4. Chai, H.; Li, Y.; Song, R.; Zhang, G.; Zhang, Q.; Liu, S.; Hou, J.; Xin, Y.; Yuan, M.; Zhang, G.; et al. A survey of the development of quadruped robots: Joint configuration, dynamic locomotion control method and mobile manipulation approach. *Biomim. Intell. Robot.* **2022**, *2*, 100029. [CrossRef]
5. Jiang, Z.; Zhang, H.; Xu, J.; Tian, G.; Rong, X. Real-time target detection and tracking system based on stereo camera for quadruped robots. In Proceedings of the 2019 Chinese Control And Decision Conference (CCDC), Nanchang, China, 3–5 June 2019; pp. 2949–2954.
6. Xu, J.; Wu, X.; Li, R.; Wang, X. Obstacle Overcoming Gait Design for Quadruped Robot with Vision and Tactile Sensing Feedback. In Proceedings of the 2021 4th International Conference on Robotics, Control and Automation Engineering (RCAE), Guangzhou, China, 26–28 November 2021; pp. 272–277.
7. Kim, D.; Lee, J.; Ahn, J.; Campbell, O.; Hwang, H.; Sentis, L. Computationally-Robust and Efficient Prioritized Whole-Body Controller with Contact Constraints. In Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Madrid, Spain, 1–5 October 2018.
8. Di Carlo, J.; Wensing, P.M.; Katz, B.; Bledt, G.; Kim, S. Dynamic Locomotion in the MIT Cheetah 3 Through Convex Model-Predictive Control. In Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Madrid, Spain, 1–5 October 2018; pp. 1–9.
9. Henze, B.; Roa, M.A.; Ott, C. Passivity-based whole-body balancing for torque-controlled humanoid robots in multi-contact scenarios. *Int. J. Robot. Res.* **2016**, *35*, 0278364916653815. [CrossRef]
10. Righetti, L.; Schaal, S. Quadratic programming for inverse dynamics with optimal distribution of contact forces. In Proceedings of the 12th International Conference on Humanoid Robots (Humanoids), Osaka, Japan, 29 November–1 December 2012; pp. 538–543.
11. Kim, D.; Di Carlo, J.; Katz, B.; Bledt, G.; Kim, S. Highly Dynamic Quadruped Locomotion via Whole-Body Impulse Control and Model Predictive Control. *arXiv* **2019**, arXiv:1909.06586.
12. Hoeller, D.; Rudin, N.; Sako, D.; Hutter, M. ANYmal parkour: Learning agile navigation for quadrupedal robots. *Sci. Robot.* **2024**, *9*, eadi7566. [CrossRef] [PubMed]
13. Allred, C.; Russell, M.; Harper, M.; Pusey, J. Improving Methods for Multi-Terrain Classification Beyond Visual Perception. In Proceedings of the Fifth IEEE International Conference on Robotic Computing (IRC), Taichung, Taiwan, 15–17 November 2021; pp. 96–99. [CrossRef]
14. Michel, A.N.; Hu, B. Towards a stability theory of general hybrid dynamical systems. *Automatica* **1999**, *35*, 371–384. [CrossRef]
15. Lygeros, J.; Johansson, K.H.; Simic, S.N.; Zhang, J.; Sastry, S.S. Dynamical properties of hybrid automata. *IEEE Trans. Autom. Control* **2003**, *48*, 2–17. [CrossRef]
16. Yang, S.; Zhang, Z.; Bokser, B.; Manchester, Z. Multi-IMU Proprioceptive Odometry for Legged Robots. In Proceedings of the 2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Detroit, MI, USA, 1–5 October 2023; pp. 774–779.
17. Margolis, G.B.; Yang, G.; Paigwar, K.; Chen, T.; Agrawal, P. Rapid locomotion via reinforcement learning. *Int. J. Robot. Res.* **2022**, *2023*, 02783649231224053.
18. Margolis, G.B.; Agrawal, P. Walk these ways: Tuning robot control for generalization with multiplicity of behavior. In Proceedings of the Conference on Robot Learning, PMLR, Atlanta, GA, USA, 6–9 November 2023; pp. 22–31.
19. Schneider, L.; Frey, J.; Miki, T.; Hutter, M. Learning risk-aware quadrupedal locomotion using distributional reinforcement learning. *arXiv* **2023**, arXiv:2309.14246.
20. He, T.; Zhang, C.; Xiao, W.; He, G.; Liu, C.; Shi, G. Agile but safe: Learning collision-free high-speed legged locomotion. *arXiv* **2024**, arXiv:2401.17583.
21. Maran, D.; Metelli, A.M.; Papini, M.; Restelli, M. No-regret reinforcement learning in smooth MDPs. *arXiv* **2024**, arXiv:2402.03792.
22. Ji, X.; Li, G. Regret-optimal model-free reinforcement learning for discounted MDPs with short burn-in time. In Proceedings of the 37th International Conference on Neural Information Processing Systems, 2023, New Orleans, LA, USA, 10–16 December 2023; Volume 36, pp. 80674–80689.
23. Ghai, U.; Gupta, A.; Xia, W.; Singh, K.; Hazan, E. Online nonstochastic model-free reinforcement learning. In Proceedings of the 37th International Conference on Neural Information Processing Systems (NeurIPS 2023), New Orleans, LA, USA, 10–16 December 2023; pp. 23362–23388. [CrossRef]

24. Wachi, A.; Shen, X.; Sui, Y. A survey of constraint formulations in safe reinforcement learning. *arXiv* **2024**, arXiv:2402.02025.
25. Wachi, A.; Hashimoto, W.; Shen, X.; Hashimoto, K. Safe exploration in reinforcement learning: A generalized formulation and algorithms. In Proceedings of the 37th International Conference on Neural Information Processing Systems (NeurIPS 2023), New Orleans, LA, USA, 10–16 December 2023; pp. 29252–29272.
26. Zhang, Q.; Leng, S.; Ma, X.; Liu, Q.; Wang, X.; Liang, B.; Liu, Y.; Yang, J. CVaR-constrained policy optimization for safe reinforcement learning. *IEEE Trans. Neural Netw. Learn. Syst.* **2024**, *36*, 830–841. [CrossRef] [PubMed]
27. Wang, H.; Qin, J.; Kan, Z. Shielded planning guided data-efficient and safe reinforcement learning. *IEEE Trans. Neural Netw. Learn. Syst.* **2024**, *36*, 3808–3819. [CrossRef] [PubMed]
28. Wachi, A.; Hashimoto, W.; Hashimoto, K. Long-term safe reinforcement learning with binary feedback. *arXiv* **2024**, arXiv:2401.03786. [CrossRef]
29. Wei, H.; Liu, X.; Ying, L. Safe reinforcement learning with instantaneous constraints: The role of aggressive exploration. In Proceedings of the AAAI Conference on Artificial Intelligence, Vancouver, BC, Canada, 20–27 February 2024; Volume 38, pp. 21708–21716.
30. Ma, H.; Liu, C.; Li, S.E.; Zheng, S.; Sun, W.; Chen, J. Learn zero-constraint-violation safe policy in model-free constrained reinforcement learning. *IEEE Trans. Neural Netw. Learn. Syst.* **2024**, *2024*, 2327–2341. [CrossRef] [PubMed]
31. Wang, X.; Knoedler, L.; Mathiesen, F.B.; Alonso-Mora, J. Simultaneous synthesis and verification of neural control barrier functions through branch-and-bound verification-in-the-loop training. *arXiv* **2023**, arXiv:2311.10438.
32. Zhang, X.; Peng, Y.; Luo, B.; Pan, W.; Xu, X.; Xie, H. Model-based safe reinforcement learning with time-varying constraints: Applications to intelligent vehicles. *IEEE Trans. Ind. Electron.* **2024**, *71*, 12744–12753. [CrossRef]
33. Ye, C.; He, J.; Gu, Q.; Zhang, T. Towards robust model-based reinforcement learning against adversarial corruption. *arXiv* **2024**, arXiv:2402.08991.
34. Lokhande, S.; Dailey, J.; Liu, Y.; Connolly, S.; Xu, H. A novel explainable AI-based situation recognition for autonomous robots with partial unlabeled data. In *Proceedings of the SPIE 12546, Sensors and Systems for Space Applications XVI*; SPIE: Bellingham, WA, USA, 2023; p. 1254606.
35. Lokhande, S.; Dailey, J.; Liu, Y.; Connolly, S.; Xu, H. An intelligent mission-oriented and situation aware quadruped robot: A novel embodied explainable AI approach. In *Proceedings of the SPIE 12549, Unmanned Systems Technology XXV*; SPIE: Bellingham, WA, USA, 2023; p. 125490E.
36. Chen, L. Optimized Foothold Planning and Posture Searching for Energy-Efficient Quadruped Locomotion Over Challenging Terrains. In Proceedings of the IEEE International Conference on Robotics and Automation (ICRA), Paris, France, 31 May–4 June 2020; pp. 399–405.
37. Norby, J.; Yang, Y.; Tajbakhsh, A.; Ren, J.; Yim, J.K.; Stutt, A.; Yu, Q.; Flowers, N.; Johnson, A.M. Quad-SDK: Full Stack Software Framework for Agile Quadrupedal Locomotion. In Proceedings of the ICRA Workshop on Legged Robots, Philadelphia, PA, USA, 23–27 May 2022.
38. Kumar, A.; Fu, Z.; Pathak, D.; Malik, J. RMA: Rapid Motor Adaptation for legged robots. In Proceedings of the Robotics: Science and Systems, Virtual, 12–16 July 2020.

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